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Determinants — Practice Worksheet

Class 11 Mathematics · 9Education

Name: ______________________ Date: ______________ Section: __________

Evaluate each determinant. Where a question says "Solve", find the value(s) of x. Show your working.

Part A — 2×2 Determinants

1. $\begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix}$

2. $\begin{vmatrix} 5 & 7 \\ 2 & 3 \end{vmatrix}$

3. $\begin{vmatrix} 6 & 4 \\ 3 & 2 \end{vmatrix}$

4. $\begin{vmatrix} -2 & 3 \\ 4 & 1 \end{vmatrix}$

5. $\begin{vmatrix} 7 & -3 \\ -2 & 5 \end{vmatrix}$

6. $\begin{vmatrix} 0 & 5 \\ 8 & 0 \end{vmatrix}$

7. $\begin{vmatrix} -4 & -2 \\ 3 & -1 \end{vmatrix}$

8. $\begin{vmatrix} 8 & 0 \\ 0 & 8 \end{vmatrix}$

9. $\begin{vmatrix} 2 & 5 \\ -1 & 3 \end{vmatrix}$

10. $\begin{vmatrix} 10 & 6 \\ 5 & 3 \end{vmatrix}$

11. $\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$

12. $\begin{vmatrix} a & b \\ b & a \end{vmatrix}$

13. Solve: $\begin{vmatrix} x & 4 \\ 1 & x \end{vmatrix} = 0$

14. $\begin{vmatrix} 12 & 8 \\ 9 & 6 \end{vmatrix}$

15. Solve: $\begin{vmatrix} x+1 & 3 \\ 2 & x \end{vmatrix} = 0$

Part B — 3×3 Determinants

16. $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

17. $\begin{vmatrix} 2 & 0 & 1 \\ 3 & 1 & 2 \\ 1 & 0 & 3 \end{vmatrix}$

18. $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$

19. $\begin{vmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{vmatrix}$

20. $\begin{vmatrix} 3 & 1 & 2 \\ 1 & 0 & 4 \\ 2 & 1 & 3 \end{vmatrix}$

21. $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 5 & 2 \end{vmatrix}$

22. $\begin{vmatrix} 2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & 2 \end{vmatrix}$

23. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{vmatrix}$

24. $\begin{vmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 2 & 3 & 0 \end{vmatrix}$

25. $\begin{vmatrix} 5 & 2 & 1 \\ 3 & 1 & 4 \\ 2 & 0 & 3 \end{vmatrix}$

26. $\begin{vmatrix} 2 & -1 & 3 \\ 1 & 4 & -2 \\ 3 & 2 & 1 \end{vmatrix}$

27. $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{vmatrix}$

28. $\begin{vmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{vmatrix}$

29. $\begin{vmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ 4 & 1 & 1 \end{vmatrix}$

30. $\begin{vmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 5 & 7 & 9 \end{vmatrix}$

31. $\begin{vmatrix} 4 & 2 & 1 \\ 6 & 3 & 2 \\ 8 & 4 & 3 \end{vmatrix}$

32. $\begin{vmatrix} 1 & 3 & 2 \\ 4 & 1 & 3 \\ 2 & 5 & 2 \end{vmatrix}$

33. $\begin{vmatrix} 2 & 0 & 0 \\ 4 & 3 & 0 \\ 5 & 6 & 1 \end{vmatrix}$

34. $\begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{vmatrix}$

35. $\begin{vmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 1 \end{vmatrix}$

36. $\begin{vmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 2 & 1 & 0 \end{vmatrix}$

37. $\begin{vmatrix} 4 & 1 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{vmatrix}$

38. $\begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{vmatrix}$

39. $\begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix}$

40. Solve: $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix} = 0$

41. $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{vmatrix}$

42. $\begin{vmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{vmatrix}$

43. $\begin{vmatrix} 0 & 2 & 0 \\ 3 & 0 & 4 \\ 0 & 5 & 0 \end{vmatrix}$

44. $\begin{vmatrix} 2 & 3 & 1 \\ 4 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix}$

45. $\begin{vmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \\ 2 & 3 & 1 \end{vmatrix}$

46. $\begin{vmatrix} 5 & 0 & 1 \\ 1 & 3 & 0 \\ 0 & 2 & 4 \end{vmatrix}$

47. $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 1 & 4 \end{vmatrix}$

48. $\begin{vmatrix} 2 & -3 & 1 \\ 4 & 1 & -2 \\ 1 & 2 & 3 \end{vmatrix}$

49. $\begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 5 & 0 & 6 \end{vmatrix}$

50. $\begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix}$

51. $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix}$

52. $\begin{vmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{vmatrix}$

53. $\begin{vmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 4 & 2 \end{vmatrix}$

54. $\begin{vmatrix} 3 & 2 & -1 \\ 1 & -1 & 2 \\ 2 & 1 & 3 \end{vmatrix}$

55. $\begin{vmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{vmatrix}$

End of worksheet. Determinants, Class 11 Mathematics.

Answers (for self-checking)

Check your final answers below. If yours doesn't match, re-do the expansion — don't just copy.

1. 10
2. 1
3. 0
4. -14
5. 29
6. -40
7. 10
8. 64
9. 11
10. 0
11. 1
12. $a^2 - b^2$
13. $x = \pm 2$
14. 0
15. $x = 2, -3$
16. 0
17. 5
18. 1
19. 40
20. -5
21. 0
22. -8
23. 1
24. 12
25. 11
26. -7
27. 1
28. 30
29. 5
30. 0
31. 0
32. 17
33. 6
34. 16
35. 27
36. -15
37. 24
38. 5
39. 4
40. $x = 1, -2$
41. -3
42. 20
43. 0
44. -25
45. 28
46. 62
47. 0
48. 63
49. 58
50. abc
51. -18
52. 32
53. -6
54. -16
55. 4